

Instructions for preparing the solution script:

- Write your name, ID#, and Section number clearly in the very front page.
- Write all answers sequentially.
- Start answering a question (not the part of the question) from the top of a new page.
- Write legibly and in orderly fashion maintaining all mathematical norms and rules.
- Start working right away. There is no late submission policy.

1. [2 marks] Define an over-determined system.
2. [8 marks] Consider a set of four data points: $f(0) = 3$, $f(4) = -2$, $f(-1) = 2$, $f(1) = 1$. Find the best fit polynomial of degree 2 by using the Discrete-square approximation.